AP PRECALCULUS

Unit 1 • Lesson 1.3 • Notes (filled in)

Topic 1.3 • Rates of Change in Linear and Quadratic Functions

Notes (filled in)

Learning goals

- Recognize linear functions by their constant rate of change.
- Recognize quadratic functions by their constant second differences, and find the quadratic from a table.
- Connect the sign of the leading coefficient to concavity and to the vertex.

Vocabulary. linear function, slope, quadratic function, first differences, second differences, vertex, axis of symmetry.

1. Linear functions

Key idea

A linear function $f(x) = mx + b$ has the __same__ average rate of change, equal to __m (the slope)__, over every interval. In a table with equally spaced inputs, the outputs change by a __constant amount__.

2. Quadratic functions

Key idea

For a quadratic $f(x) = ax^2 + bx + c$, the average rates of change over consecutive equal-length intervals change at a __constant__ rate. With equally spaced inputs, the __second differences__ are constant. When the input step is 1, the second differences equal __$2a$__.

Example 1 • Classify each table

x	0	1	2	3
A	4	7	10	13
B	1	2	5	10
C	1	2	4	8

A: differences $3, 3, 3$: linear. B: differences $1, 3, 5$, second differences $2, 2$: quadratic ($y = x^2 + 1$). C: differences $1, 2, 4$, second differences $1, 2$: neither linear nor quadratic (it doubles: exponential).

Example 2 • Finding the quadratic

x	0	1	2	3	4
y	3	1	-3	-9	-17

Show that the data are quadratic and find the function.

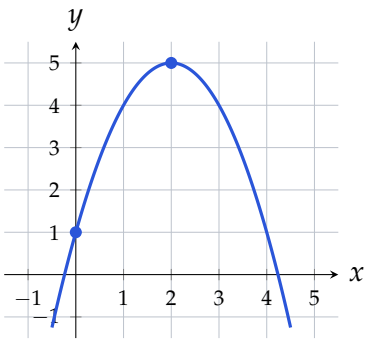
First differences: $-2, -4, -6, -8$. Second differences: $-2, -2, -2$. Quadratic with $2a = -2$, so $a = -1$.
$c = y(0) = 3$. Using $x = 1$: $-1 + b + 3 = 1$, so $b = -1$. $\quad y = -x^2 - x + 3$.

3. Concavity and the vertex

Key idea

If $a > 0$, the rate of change is __increasing__: the parabola is concave __up__ and has a __minimum__ at the vertex.
If $a < 0$, it is concave __down__ with a __maximum__. The vertex is at $x =$ __$-\frac{b}{2a}$__; there the rate of change changes sign.

Example 3 • From a graph



Write an equation for this parabola and compute the average rates of change on $[0,1]$, $[1,2]$, $[2,3]$.

Vertex $(2,5)$: $f(x) = a(x-2)^2 + 5$. Through $(0,1)$: $4a + 5 = 1$, $a = -1$. So $f(x) = -(x-2)^2 + 5$. $f(0) = 1$, $f(1) = 4$, $f(2) = 5$, $f(3) = 4$: rates 3, 1, -1. They decrease by 2 each step (concave down).

Common mistakes

- Using differences when the inputs are not equally spaced.
- Thinking the second difference equals a; it equals $2a$ when the step is 1.

Check your understanding

1. Outputs $2, 5, 10, 17$ at $x = 0, 1, 2, 3$ come from a __quadratic__ function.
2. Second differences of 6 (step 1) mean $a =$ __3__, and the graph is concave __up__.